

BST Physics Curriculum Vol 10 Chapter 2 — Complex Analysis + Holomorphic Functions v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 10 primary]

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Vol 10 Chapter 2 — Complex Analysis + Holomorphic Functions

Chapter motivation

Standard graduate complex analysis treats: holomorphic functions on $\mathbb{C}$; Cauchy's integral formula + theorem; residue theorem; conformal mappings (Riemann mapping theorem); Riemann surfaces; meromorphic functions + analytic continuation. Standard texts: Ahlfors + Conway + Stein-Shakarchi.

BST cross-link: the substrate $D_{IV^5} \subset \mathbb{C}^5$ is a bounded holomorphic substrate (type IV bounded Hermitian symmetric domain per Cartan 1894 + Hua 1958, Vol 11 Ch 1). The Bergman space $H^2(D_{IV^5})$ consists of holomorphic L^2 -functions on D_{IV^5} ; the Bergman reproducing kernel $K_B(z, \bar{w})$ (Bergman 1922 + Faraut-Koranyi 1994; Vol 11 Ch 2) is holomorphic in z + antiholomorphic in $\bar{w}$. Substrate physics IS complex-analytic at substrate scale.

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: Standard complex analysis (Cauchy + residues + conformal maps + Riemann surfaces); BST cross-link: $D_{IV^5} \subset \mathbb{C}^5$ is bounded holomorphic substrate; Bergman H^2 = holomorphic L^2 -functions on D_{IV^5} ; Bergman kernel $K_B(z, \bar{w})$ holomorphic in z , antiholomorphic in $\bar{w}$.

Level 2 (graduate-physicist): Standard complex analysis: holomorphic functions $f: U \subset \mathbb{C} \rightarrow \mathbb{C}$ satisfying Cauchy-Riemann equations; Cauchy's integral formula $f(z) = (1/(2\pi i)) \oint f(\zeta)/(\zeta-z) d\zeta$; residue theorem $\oint f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$; conformal mappings preserve angles + orientations; Riemann mapping theorem (simply-connected domains biholomorphic to disk); Riemann surfaces for multi-valued functions (logarithm, square root); meromorphic functions = holomorphic except for poles; analytic continuation extends function maximally. BST substrate

cross-link: $D_{IV^5} \subset \mathbb{C}^5$ is type IV bounded Hermitian symmetric domain (Cartan 1894 + Hua 1958 + Vol 11 Ch 1), bounded by $1 - 2(\bar{z} \cdot z) + |z \cdot z|^2 > 0 + |z \cdot z| < 1$. Bergman space $H^2(D_{IV^5})$ (Vol 1 Ch 2 + Vol 11 Ch 2) = holomorphic L^2 -functions; Bergman reproducing kernel $K_B(z, \bar{w}) = c_{FK} \cdot h(z, \bar{w})^{-(g/\text{rank})} = c_{FK} \cdot h(z, \bar{w})^{-(7/2)}$ per Faraut-Koranyi 1994; $c_{FK} \cdot \pi^{((g+\text{rank})/\text{rank})} = (N_C \cdot n_C)^2 = 225$ EXACT per T2442 Strong-Uniqueness C13. Substrate-cartography: substrate physics IS complex-analytic at substrate scale; holomorphic structure on D_{IV^5} underlies all BST observables; substrate operators act as multiplication/differentiation on holomorphic Bergman H^2 functions. Cauchy integral formula on D_{IV^5} generalizes to reproducing-kernel reproducing property $f(w) = \int_{D_{IV^5}} K_B(z, \bar{w}) f(z) dV(z)$; standard Cauchy formula recovered at planar boundary limit.

Level 3 (5th-grader accessible): Complex analysis = calculus with complex numbers. Used throughout BST since the substrate D_{IV^5} is a bounded 5-complex-dimensional space. The Bergman kernel (a special holomorphic function on the substrate) is the substrate's reproducing-kernel function — knowing this kernel + a function's values on the boundary lets you reconstruct the function inside the substrate. Standard complex-analysis tools (Cauchy's integral formula, residue theorem, conformal mapping) all generalize to BST's substrate framework.

Section 2.1 — Standard Complex Analysis

Holomorphic $f: U \subset \mathbb{C} \rightarrow \mathbb{C}$ satisfies Cauchy-Riemann $\partial f / \partial x = \partial f / \partial y \cdot i$.

Cauchy's integral formula: $f(z) = (1/(2\pi i)) \oint f(\zeta) / (\zeta - z) d\zeta$ for f holomorphic in disk containing z .

Residue theorem: $\oint_C f(z) dz = 2\pi i \cdot \sum \text{Res}(f, z_k)$ for f meromorphic inside contour C with poles z_k .

Conformal mappings + Riemann mapping theorem: every simply-connected proper domain in $\mathbb{C}$ is biholomorphic to the unit disk.

Section 2.2 — D_{IV^5} Bounded Holomorphic Substrate (Cartan 1894 + Hua 1958)

Per Vol 0 Ch 1 + Vol 11 Ch 1: $D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$ type IV bounded Hermitian symmetric domain of complex dimension $n_C = 5$, rank = 2.

Cartan 1894 + Hua 1958 explicit realization: $D_{IV^5} \subset \mathbb{C}^5$ as set of $z = (z_1, \dots, z_5) \in \mathbb{C}^5$ satisfying $1 - 2(\bar{z} \cdot z) + |z \cdot z|^2 > 0 + |z \cdot z| < 1$.

Section 2.3 — Bergman Space $H^2(D_{IV^5})$ (Vol 11 Ch 2)

Bergman 1922: Bergman space $H^2(D_{IV^5})$ of holomorphic L^2 -functions on D_{IV^5} ; reproducing kernel $K_B(z, \bar{w})$ such that $f(w) = \int_{D_{IV^5}} K_B(z, \bar{w}) f(z) dV(z)$ for $f \in H^2$.

Faraut-Koranyi 1994 explicit formula on type IV: $K_B(z, \bar{w}) = c_{FK} \cdot h(z, \bar{w})^{-(g/\text{rank})} = c_{FK} \cdot h(z, \bar{w})^{-7/2}$ where $h(z, \bar{w})$ is the generic norm.

T2442 Strong-Uniqueness C13: $c_{FK} \cdot \pi^{((g+\text{rank})/\text{rank})} = (N_c \cdot n_C)^2 = 225$ EXACT.

Section 2.4 — Substrate-Cartography Reading

Substrate physics IS complex-analytic at substrate scale. Holomorphic structure on D_{IV^5} underlies all BST observables. Substrate operators (Vol 1 Ch 6) act as multiplication/differentiation on holomorphic Bergman H^2 functions: - Position $M_z f(z) = z \cdot f(z)$ (multiplication) - Momentum $P_z f(z) = \partial_z f(z)$ (Wirtinger derivative) - Cross-link Vol 5 Ch 2

Section 2.5 — Cauchy → Reproducing Kernel Generalization

Standard Cauchy integral formula on planar disk: $f(z) = (1/(2\pi i)) \oint f(\zeta)/(\zeta - z) d\zeta$

BST generalization on D_{IV^5} : $f(w) = \int_{D_{IV^5}} K_B(z, \bar{w}) f(z) dV(z)$

Standard Cauchy formula recovered at planar boundary limit (Vol 0 Ch 4 §4.6 Shilov boundary).

Section 2.6 — Honest scope + Connection

What this chapter establishes: standard complex analysis machinery + explicit BST substrate cross-link to D_{IV^5} bounded holomorphic substrate + Bergman reproducing kernel framework.

Open scope: - Quantitative Cauchy → reproducing-kernel reduction at conformal boundary (multi-week) - Higher-dim Riemann-surface generalization on D_{IV^5} - Cross-link to Vol 10 Ch 5 Fourier analysis (Wallach K-type)

Connection: - Vol 0 Ch 1 D_{IV^5} APG - Vol 1 Ch 2 Substrate Hilbert Space - Vol 11 Ch 2 Bergman Reproducing Kernels (detailed treatment) - Vol 5 Ch 2 Position + Momentum ($M_z + P_z$ on Bergman H^2)

— Lyra, Vol 10 Ch 2 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT